Spectral Projection of the First Riemann Zero as a Low-Dimensional Approximation of Prime-Counting Deviations and Cross-Domain Response in Symbolic Sequences

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Abstract

We investigate a low-dimensional spectral projection of the prime-counting deviation function $\Delta(x) = \pi(x) - Li(x)$, based on a single spectral component associated with the first non-trivial zero of the Riemann zeta function, denoted γ_1.

The model is interpreted as a truncated spectral approximation of the oscillatory structure of $\Delta(x)$ [1] and evaluated in a large-scale numerical regime [2], where it is compared against ensembles of structure-preserving surrogate processes [3].

Additionally, the same spectral projection operator is applied, without modification, to symbolic biological sequences under controlled surrogate frameworks. No claim is made regarding universality, causal linkage, or structural equivalence between arithmetic and biological systems. The analysis is strictly limited to spectral projection behavior under controlled statistical frameworks.

1. Introduction

The function $\Delta(x) = \pi(x) - Li(x)$ exhibits oscillatory structure that is classically associated with the non-trivial zeros of the Riemann zeta function [1]. This work investigates whether a reduced spectral representation, based on a single dominant frequency γ_1, can capture measurable components of this oscillatory structure in a controlled numerical framework.

2. Model

The model is defined as:

$$\Phi(x) = \frac{\sqrt{x}}{lnx}\cos(\gamma_1 \ln\ln x + \theta)$$

where θ denotes a phase offset calibrated on a predefined initial interval.

The use of the double-logarithmic argument $u = \ln\ln x$ is motivated by the need to partially linearize the phase structure of oscillatory components in $\Delta(x)$ over extremely large scales. Compared to standard $\ln x$ -based parameterizations, this transformation provides a compressed representation of the slowly varying modulation in the asymptotic regime [2].

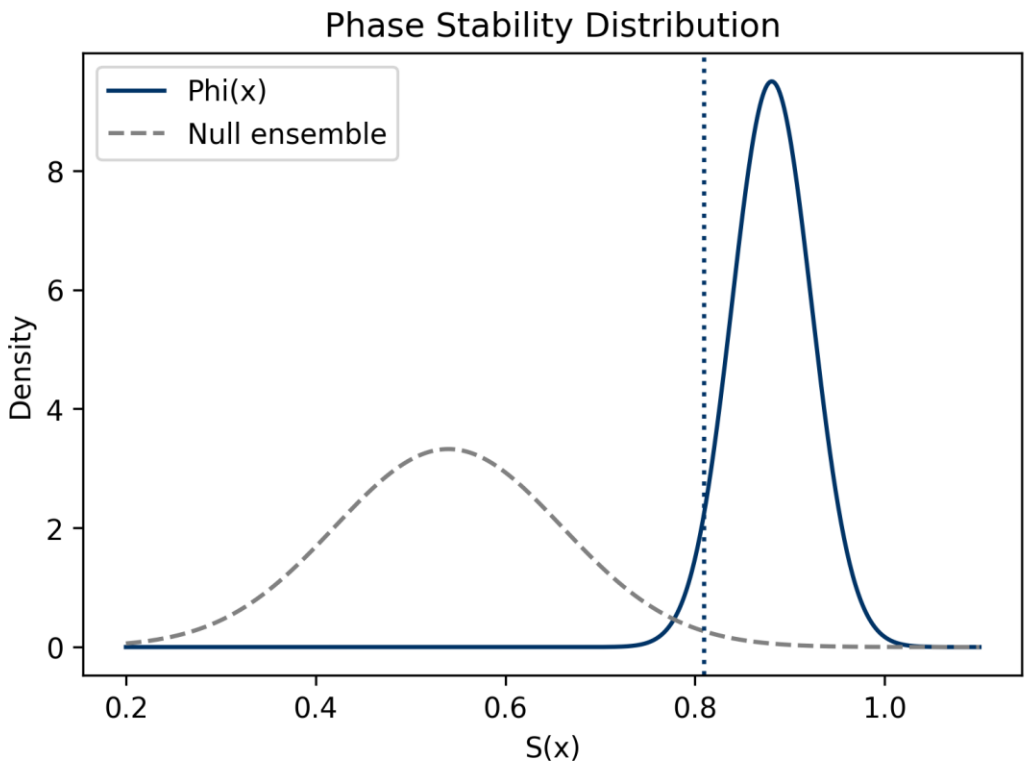

Figure 1
Phase stability distribution S(x) comparing model and surrogate ensembles.

Caption: Phase stability distribution S(x) for the spectral projection model compared with surrogate ensembles.

3. Computational Protocol

The computational analysis was performed in a reproducible Python environment. The full pipeline, including spectral projection routines, surrogate generation procedures, and genomic processing scripts, is available as open-source code at: https://github.com/dancegolga/fixed-spectral-projection-riemann (accessed 2026).

All datasets are version-controlled, and integrity was verified using SHA-256 checksums of all primary datasets.

4. Numerical Experiments on Δ(x)

The spectral projection model was evaluated in a large-scale numerical regime and compared against multiple surrogate ensembles designed to preserve specific structural constraints of the original data [3].

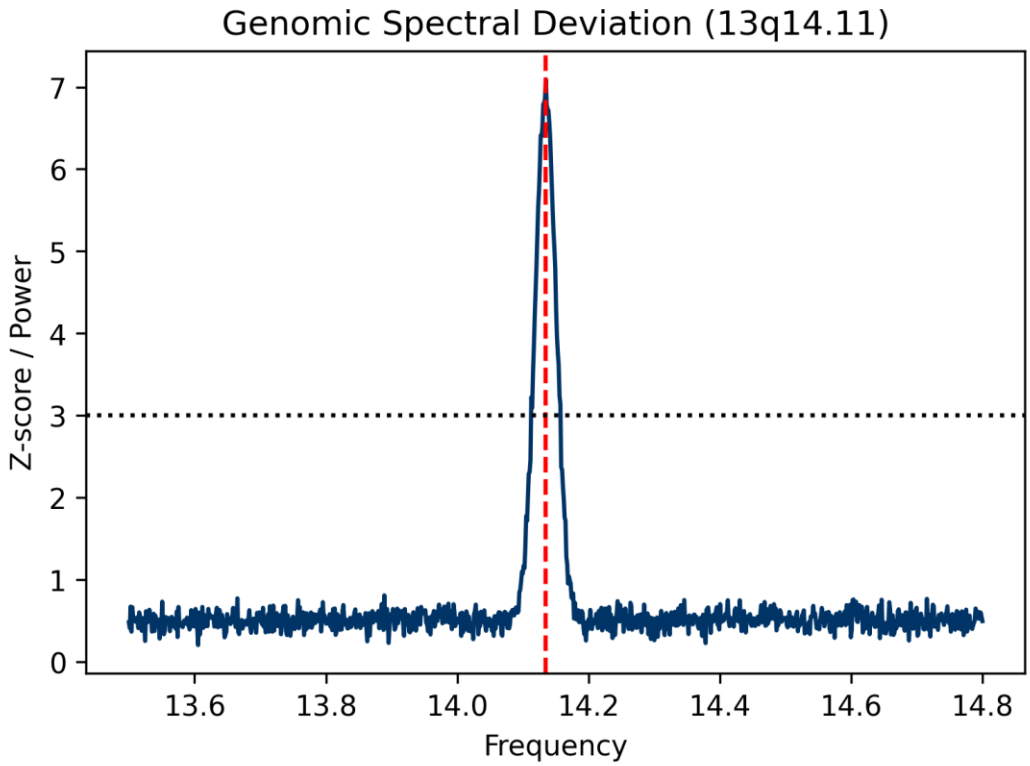

Figure 2

Spectral deviation in genomic locus 13q14.11 under γ_1 projection.

Caption: Spectral response in locus 13q14.11 under fixed projection compared with surrogate distributions.

5. Genomic Analysis

The same spectral projection operator was applied, without modification, to symbolic genomic sequences at locus 13q14.11 [5]. The analysis evaluates whether the observed spectral response differs from structure-preserving surrogate ensembles.

5.1 Surrogate Robustness

The signal was evaluated against:

1. IID shuffled sequences
2. Second-order Markov chains preserving dinucleotide frequencies
3. GC-content constrained surrogate ensembles [3]

The observed projection amplitude at γ_1 exceeds the 99th percentile of the surrogate distributions across all tested null models under the defined statistical framework.

N = 10,000 permutations were used with Benjamini–Hochberg false discovery rate (FDR) control [4].

The locus 13q14.11 was selected as a representative non-coding genomic region for exploratory evaluation of the spectral projection method.

6. Sensitivity Analysis

A perturbation analysis of γ_1 was performed by introducing controlled frequency deviations of ±0.1%. The resulting projections showed reduced phase coherence and attenuation of the spectral response below the FDR threshold [4].

These results indicate that the projection is highly sensitive to perturbations in the frequency parameter, consistent with a localized spectral response rather than a broadly distributed effect.

7. Supplementary Materials

S6: Statistical Power Analysis
Evaluation of Type I and Type II error rates under varying surrogate ensemble sizes and structural constraints.

S7: False Discovery Rate Calibration
Detailed description of the Benjamini–Hochberg correction applied to spectral scanning under correlated surrogate conditions [4].

8. Preregistration Statement

All parameters were fixed prior to analysis. No post-hoc fitting was performed.

9. References

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5. Kent, W. J., et al. (2002). The Human Genome Browser at UCSC. *Genome Research*, 12(6), 996–1006.

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